CS 411 Lab 1

Beverage Operating System

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29 September 2026

Version 2

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# 1 Introduction

This section outlines the introduction to the societal problem, needed solutions, and a brief overview to the product.

## 1.1 Societal Problem

The beverage manufacturing industry relies on spreadsheets, email chains, phone calls, and disconnected software systems to coordinate production activities. As beverage companies grow, these manual processes create communication delays, approval bottlenecks, and operational inefficiencies. The beverage packaging market was valued at approximately $157.7 billion in 2023 and is projected to reach $222.1 billion by 2030 (Grand View Research, 2024). Beverage manufacturers need a centralized system that improves collaboration across suppliers, production teams, logistics providers, and clients. The complexity of beverage manufacturing and packaging investments can sometimes create tension and miscommunications, as illustrated in Figure 1. This increases the need for coordinated, scalable information systems that help organizations manage production activity across multiple stakeholders (Zimmerman, 2026).

## 1.2 Problem and Needed Solution Characteristics

Centralized workflow management organizes operational activities within one platform and reduces dependence on scattered spreadsheets and email. Real-time visibility into production activities allows authorized stakeholders to view current requests, approvals, schedules, production progress, and delivery status. Automated approval tracking records responsibility and status while reducing delays caused by repeated manual follow-up.

The production scheduling helps coordinates milestones and helps teams recognize delays before they affect delivery. The logistics coordination connects production completion, shipping activity, and delivery information. The document management system stores operational files in an organized location where authorized users can locate the current version. The reporting and dashboard capabilities turns operational data into useful information for oversight and decision making. The secure role-based user access limits users to the information and actions appropriate to their responsibilities. The application’s scalability will support business growth by supporting additional users, customers, workflows, and business growth without replacing the operational foundation.

## 1.3 Product Introduction

BevOS is a centralized workflow management platform designed specifically for beverage manufacturers. The system connects brands, suppliers, distributors, co-packers, logistics providers, and internal operations through a single platform. BevOS improves efficiency by reducing manual coordination, improving communication, and increasing operational visibility. A functional prototype will demonstrate how the product’s interface, services, modules, database work together to address the identified coordination problem.



*Figure 1 - Communications between different parties may experience disconnect.*

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## 2 BevOS Product Description

BevOS provides a unified operational platform that centralizes workflow management, approval tracking, production scheduling, document management, and logistics coordination.

### 2.1 Purpose

The purpose is to improve visibility, reduce manual effort, and create a scalable system that supports business growth.

### 2.2 Goals and Objectives

The main goal of BevOS is to centralize operational workflows to eliminate data fragmentation across spreadsheets and email chains.

It will improve communication, transparency, and visibility among brands, suppliers, co-packers, logistics providers, and finance teams.

The system automates and track quotes, purchase orders, approvals, and production schedules.

It aims to reduce manual effort, eliminate operational bottlenecks, and establish a scalable foundation for business expansion.

### 2.3 Intended Users

For the brands, clients, and distributors, the program can initiate product requests, track production milestones, approve artwork, and oversee project lifecycles.

For suppliers and co-packers, it can receive purchase orders, respond to quotes, and coordinate manufacturing requirements.

For logistics providers and carriers, the application can manage shipping coordination, transport tracking, and delivery updates.

For operations and finance teams, the system will monitor production schedules, manage invoicing, analyze performance metrics, and review reports.

For BevOS developer and IT maintainers, it will help manage system administration, handle bug tracking/ticket queues, and ensure platform scalability.

### 2.4 Key Product Features and Capabilities

This section outlines key products features and capabilities, including workflow management, product request, quote, approval tracking, production, logistics, documentation, communications platform, report, dash boards, and issue tracking.

### 2.4.1 Workflow Management

The workflow management system centralizes all operational workflows from initial concept to store shelves, replacing disconnected spreadsheets and long email chains with a unified process pipeline.

### 2.4.2 Product Request Management

The product request management system streamlines product and artwork requests through automated generation and approval routing, escalating to partial automation when necessary.

### 2.4.3 Quote Management

The quote system facilitates automated quote calculation and supports partial automation for supplier negotiations and distributor pricing tracking.

### 2.4.4 Approval Tracking

The approval tracking system manages tracking and routing of manual and automated approvals for artwork, purchase orders, and production schedules to eliminate operational bottlenecks.

### 2.4.5 Production Management

The production management system coordinates manufacturing activities and production scheduling seamlessly with co-packers and internal production teams.

### 2.4.6 Logistics Coordination

The logistics coordination system streamlines carrier coordination, shipping updates, and delivery tracking from manufacturing facilities to final destinations.

### 2.4.7 Document Management

The application can store, organize, and manage operational files, product specifications, and legal/technical documents in a secure central repository.

### 2.4.8 Communications Platform

The communication platform provides a centralized messaging and communication channel connecting all stakeholders to prevent lost updates and communication delays.

### 2.4.9 Reporting and Dashboards

The reporting system and dashboards delivers real-time data visualization, metrics, and multi-perspective business insights through comprehensive dashboards and reporting tools.

### 2.4.10 Issue Tracking

The issue tracker integrates internal user feedback into a structured issue tracking pipeline to identify and resolve emerging operational shortcomings.

## 2.5 Major Components (Hardware/Software)

This section outlines the hardware requirements necessary to power the solution. It details its architectural organization according to the Major Functional Component Diagram (MFCD), and specify the software components to be built.

### 2.5.1 Hardware Components

The software will be hosted on standard cloud or local server infrastructure. The application and database containers will be supported by the redundant infrastructure to ensure tolerance for partial functionality during minor network outages. End-users will be able to access the system via standard desktop and mobile web-enabled client devices.

### 2.5.2 Software Components

The software will be developed using Python as the primary programming language and Django as the core web framework. The backend data layer relies on PostgreSQL, and containerization is handled via Docker and Docker Compose. Quality assurance and CI/CD pipelines utilize Pytest for testing and GitHub Actions.

### 2.5.3 MFCD Structure and Data flow

The application has a user-facing UI where users will mainly interact with. The BevOS core services makes up the core business logic of this program. It includes quotes, communications, logistics, documents storage, reports, and database as detailed in Figure 2.

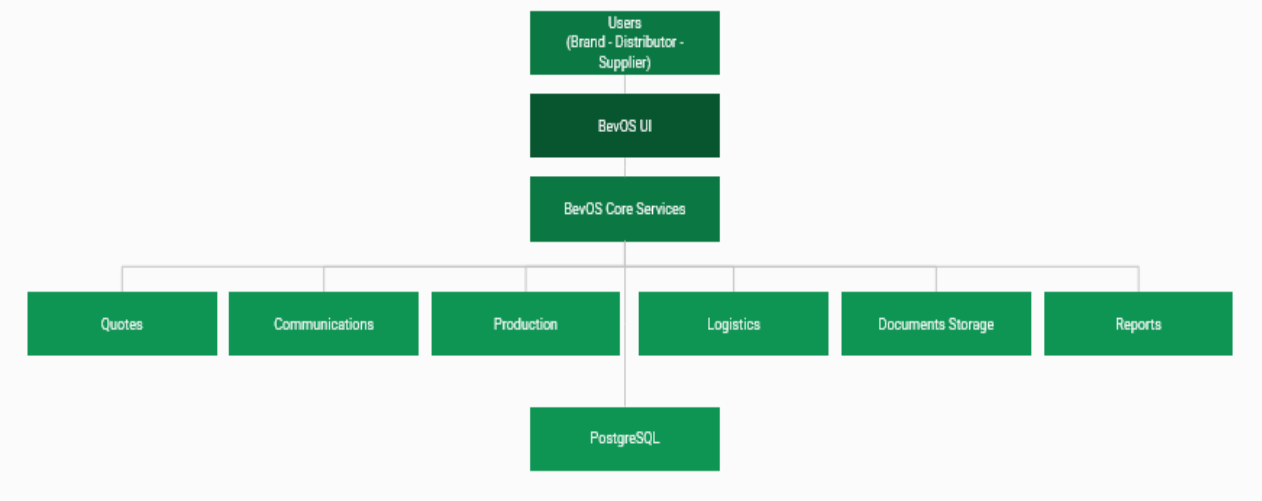



*Figure 2 – MFCD Structure*

**3 Glossary**

● Approval Workflow - A process used to review and approve operational tasks before they move to the next stage.

● BevOS - A centralized workflow-management platform designed for beverage manufacturing operations.

● Client - An organization or individual purchasing services or products from another participant in the manufacturing process.

● Co-Packer - A third-party company that manufactures and packages beverages for another company.

● Core Services - The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.

● Dashboard - A visual interface displaying operational information and performance metrics.

● Distributor - An organization responsible for delivering products from manufacturers to retailers or customers.

● ERP (Enterprise Resource Planning) - Software that integrates core business processes into a unified system.

● Logistics - The planning and coordination of transportation and delivery activities.

● MFCD (Major Functional Components Diagram) - A diagram illustrating the primary hardware and software components of the system.

● Nice Package Company - The beverage-industry client and domain partner for whom BevOS is being developed.

● PostgreSQL - An open-source relational database management system used to store application data.

● Role-Based Access Control - A security method that grants system permissions according to the user's assigned role and responsibilities.

● Single Source of Truth - A trusted, centralized set of current information used consistently by

authorized stakeholders.

● Stakeholder - A person or organization that participates in or is affected by a beverage-manufacturing workflow.

● Workflow - A sequence of business activities completed to achieve a specific objective.

## 4 References

● Grand View Research. (2024). Beverage Packaging Market Size, Share & Trends Analysis Report.

● Kennedy, T. (2026). CS410 Professional Workforce Development.

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